# Assignment 1: Imitation Learning

## 1 Analysis

1. For any $s_t$, we have

- $p_{\pi_\theta}(s_t) = \sum_{s_{t-1}} p_{\pi_\theta}(s_t|s_{t-1})p_{\pi_\theta}(s_{t-1}) = \sum_{s_{t-1}} \sum_{a_{t-1}} \pi_\theta(a_{t-1}|s_{t-1})p(s_t|s_{t-1}, a_{t-1})p_{\pi_\theta}(s_{t-1})$
- $p_{\pi^*}(s_t) = \sum_{s_{t-1}} p_{\pi^*}(s_t|s_{t-1})p_{\pi^*}(s_{t-1}) = \sum_{s_{t-1}} \sum_{a_{t-1}} \pi^*(a_{t-1}|s_{t-1})p(s_t|s_{t-1}, a_{t-1})p_{\pi^*}(s_{t-1})$

So, we can establish $\sum_{s_t} |p_{\pi_\theta}(s_t) - p_{\pi^*}(s_t)| \leq$
$\sum_{s_t} \sum_{s_{t-1}} \sum_{a_{t-1}} p(s_t|s_{t-1}, a_{t-1}) \left|\pi_\theta(a_{t-1}|s_{t-1})p_{\pi_\theta}(s_{t-1}) - \pi^*(a_{t-1}|s_{t-1})p_{\pi^*}(s_{t-1})\right|$

For each $s_{t-1}$ and $a_{t-1}$, rewrite $p_{\pi_\theta}(s_{t-1})$ as the $p_{\pi_\theta}(s_{t-1}) - p_{\pi^*}(s_{t-1}) + p_{\pi^*}(s_{t-1})$, and we get

$$\begin{aligned}
&|\pi_\theta(a_{t-1}|s_{t-1})p_{\pi_\theta}(s_{t-1}) - \pi^*(a_{t-1}|s_{t-1})p_{\pi^*}(s_{t-1})| \\
=&|\pi_\theta(a_{t-1}|s_{t-1})(p_{\pi_\theta}(s_{t-1}) - p_{\pi^*}(s_{t-1})) + \pi_\theta(a_{t-1}|s_{t-1})p_{\pi^*}(s_{t-1}) - \pi^*(a_{t-1}|s_{t-1})p_{\pi^*}(s_{t-1})| \\
\leq&\pi_\theta(a_{t-1}|s_{t-1})|p_{\pi_\theta}(s_{t-1}) - p_{\pi^*}(s_{t-1})| + |\pi_\theta(a_{t-1}|s_{t-1}) - \pi^*(a_{t-1}|s_{t-1})|p_{\pi^*}(s_{t-1})
\end{aligned}$$

Plugging the inequality back, we get an upper bound for $\sum_{s_t} |p_{\pi_\theta}(s_t) - p_{\pi^*}(s_t)|$:

$$\sum_{s_t} \sum_{s_{t-1}} \sum_{a_{t-1}} p(s_t|s_{t-1}, a_{t-1})\pi_\theta(a_{t-1}|s_{t-1})|p_{\pi_\theta}(s_{t-1}) - p_{\pi^*}(s_{t-1})| + p(s_t|s_{t-1}, a_{t-1})|\pi_\theta(a_{t-1}|s_{t-1}) - \pi^*(a_{t-1}|s_{t-1})|p_{\pi^*}(s_{t-}$$

We can bound the two terms separately.

For the first term,

$$\begin{aligned}
&\sum_{s_t} \sum_{s_{t-1}} \sum_{a_{t-1}} p(s_t|s_{t-1}, a_{t-1})\pi_\theta(a_{t-1}|s_{t-1})|p_{\pi_\theta}(s_{t-1}) - p_{\pi^*}(s_{t-1})| \\
=&\sum_{s_{t-1}} |p_{\pi_\theta}(s_{t-1}) - p_{\pi^*}(s_{t-1})|(\sum_{s_t} \sum_{a_{t-1}} p(s_t|s_{t-1}, a_{t-1})\pi_\theta(a_{t-1}|s_{t-1})) \\
=&\sum_{s_{t-1}} |p_{\pi_\theta}(s_{t-1}) - p_{\pi^*}(s_{t-1})|(\sum_{s_t} p_{\pi_\theta}(s_t|s_{t-1})) \\
=&\sum_{s_{t-1}} |p_{\pi_\theta}(s_{t-1}) - p_{\pi^*}(s_{t-1})|
\end{aligned}$$

For the second term,

$$\begin{aligned}
&\sum_{s_t} \sum_{s_{t-1}} \sum_{a_{t-1}} p(s_t|s_{t-1}, a_{t-1})|\pi_\theta(a_{t-1}|s_{t-1}) - \pi^*(a_{t-1}|s_{t-1})|p_{\pi^*}(s_{t-1}) \\
=&\sum_{s_{t-1}} \sum_{a_{t-1}} p_{\pi^*}(s_{t-1})|\pi_\theta(a_{t-1}|s_{t-1}) - \pi^*(a_{t-1}|s_{t-1})|(\sum_{s_t} p(s_t|s_{t-1}, a_{t-1})) \\
=&\sum_{s_{t-1}} \sum_{a_{t-1}} p_{\pi^*}(s_{t-1})|\pi_\theta(a_{t-1}|s_{t-1}) - \pi^*(a_{t-1}|s_{t-1})| \\
\leq&\sum_{s_{t-1}} p_{\pi^*}(s_{t-1})(\sum_{a_{t-1}} |\pi_\theta(a_{t-1}|s_{t-1}) - \pi^*(a_{t-1}|s_{t-1})|) \\
=&\sum_{s_{t-1}} p_{\pi^*}(s_{t-1})(1 - \pi_\theta(a_{t-1} = \pi^*(s_{t-1})|s_{t-1}) + \sum_{a_{t-1} \neq \pi^*(s_{t-1})} \pi_\theta(a_{t-1}|s_{t-1})) \\
=&2\sum_{s_{t-1}} p_{\pi^*}(s_{t-1})\pi_\theta(a_{t-1} \neq \pi^*(s_{t-1})|s_{t-1}) \\
=&2\mathbb{E}_{p_{\pi^*}(s_{t-1})}[\pi_\theta(a_{t-1} \neq \pi^*(s_{t-1})|s_{t-1})]
\end{aligned}$$

Putting everything together, we get

$$
\begin{aligned}
&\sum_{s_t} |p_{\pi_\theta}(s_t) - p_{\pi^*}(s_t)| \\
\leq &\sum_{s_{t-1}} |p_{\pi_\theta}(s_{t-1}) - p_{\pi^*}(s_{t-1})| + 2\mathbb{E}_{p_{\pi^*}(s_{t-1})}[\pi_\theta(a_{t-1} \neq \pi^*(s_{t-1})|s_{t-1})] \\
\leq &\sum_{s_1} |p_{\pi_\theta}(s_1) - p_{\pi^*}(s_1)| + 2\sum_{i=1}^{t-1} \mathbb{E}_{p_{\pi^*(s_i)}}[\pi_\theta(a_i \neq \pi^*(s_i)|s_i)]
\end{aligned}
$$

Since both $p_{\pi_\theta}(s_1)$ and $p_{\pi^*}(s_1)$ equal to probability of $s_0$ under the initial distribution of the MDP, we know $\sum_{s_1} |p_{\pi_\theta}(s_1) - p_{\pi^*}(s_1)| = 0$

To bound $\sum_{i=1}^{t-1} \mathbb{E}_{p_{\pi^*(s_i)}}[\pi_\theta(a_i \neq \pi^*(s_i)|s_i)]$, we use the fact that $\frac{1}{T}\sum_{i=1}^{T} \mathbb{E}_{p_{\pi^*(s_i)}}[\pi_\theta(a_i \neq \pi^*(s_i)|s_i)] \leq \epsilon$, which implies $\sum_{i=1}^{T} \mathbb{E}_{p_{\pi^*(s_i)}}[\pi_\theta(a_i \neq \pi^*(s_i)|s_i)] \leq T\epsilon$. Therefore, $\sum_{i=1}^{t-1} \mathbb{E}_{p_{\pi^*(s_i)}}[\pi_\theta(a_i \neq \pi^*(s_i)|s_i)] \leq \sum_{i=1}^{T} \mathbb{E}_{p_{\pi^*(s_i)}}[\pi_\theta(a_i \neq \pi^*(s_i)|s_i)] \leq T\epsilon$.

Now we can conclude that $\sum_{s_t} |p_{\pi_\theta}(s_t) - p_{\pi^*}(s_t)| \leq 2T\epsilon$ for any $t$.

2. We can establish the following general bound.

$$
\begin{aligned}
J(\pi^*) - J(\pi) &= \sum_{t=1}^{T} \mathbb{E}_{p_{\pi^*}(s_t)}[r(s_t)] - \sum_{t=1}^{T} \mathbb{E}_{p_\pi(s_t)}[r(s_t)] \\
&= \sum_{t=1}^{T}\sum_{s_t}(p_{\pi^*}(s_t) - p_\pi(s_t))r(s_t) \\
&\leq \sum_{t=1}^{T}\sum_{s_t}|p_{\pi^*}(s_t) - p_\pi(s_t)| \cdot |r(s_t)| \\
&\leq R_{\max}\sum_{t=1}^{T}\sum_{s_t}|p_{\pi^*}(s_t) - p_\pi(s_t)|
\end{aligned}
$$

a. When $r(s_t) = 0$ for all $t < T$, we have $J(\pi^*) - J(\pi) \leq R_{\max}\sum_{s_T}|p_{\pi^*}(s_T) - p_\pi(s_T)| \leq 2R_{\max}T\epsilon = \mathcal{O}(T\epsilon)$

b. Generally, $J(\pi^*) - J(\pi) \leq R_{\max}\sum_{t=1}^{T}\sum_{s_t}|p_{\pi^*}(s_t) - p_\pi(s_t)| \leq R_{\max}\sum_{t=1}^{T} 2T\epsilon = 2R_{\max}T^2\epsilon = \mathcal{O}(T^2\epsilon)$

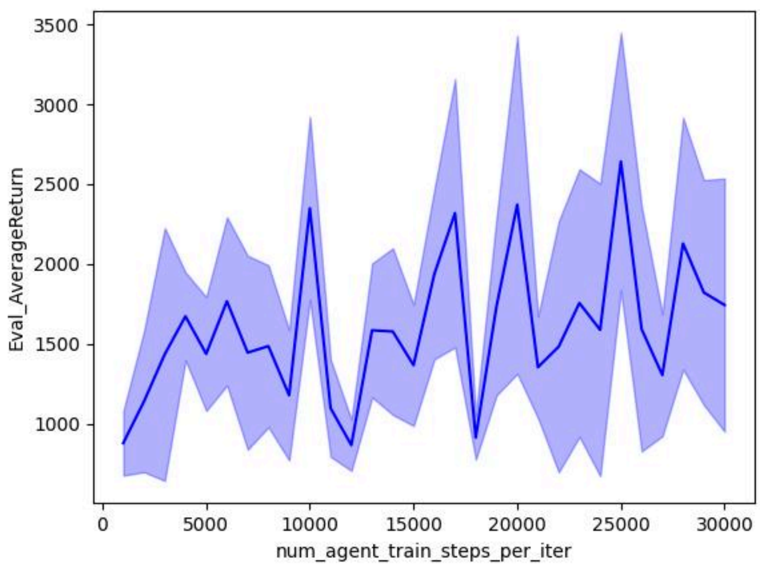
# 3 Behavior Cloning

## Default setting

| Environment | eval_batch_size | Eval_AverageEpLen | Eval_AverageReturn | Eval_StdReturn | Train_AverageReturn | Eval/Train Return Ratio |
|---|---|---|---|---|---|---|
| Ant | 5000 | 1000 | 4738.90 | 91.56 | 4681.90 | 101% |
| Walker2d | 5000 | 307 | 1287.81 | 1357.56 | 5383.31 | 24% |
| HalfCheetah | 5000 | 1000 | 3915.16 | 96.66 | 4034.80 | 97% |
| Hopper | 5000 | 317 | 879.51 | 202.26 | 3717.51 | 24% |

## Hyperparameter tuning

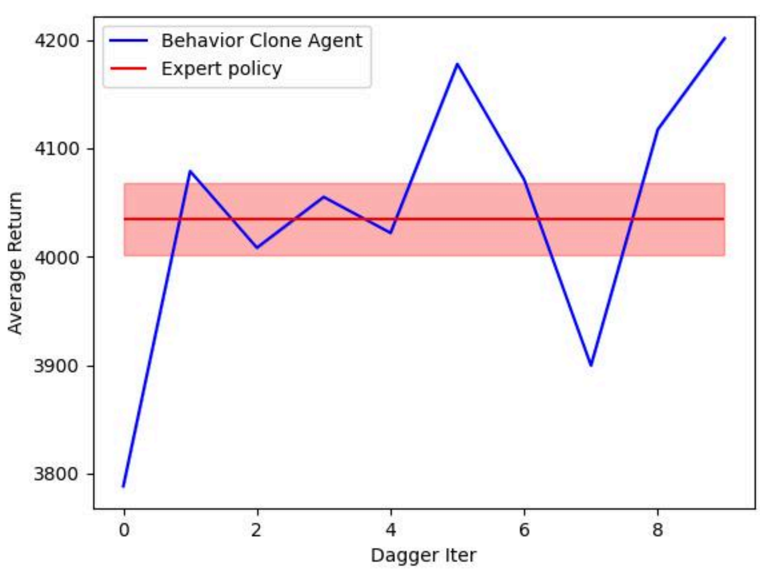
Since we only train 1000 steps under the default setting, it's likely that the model hasn't converged to the local optimal. The hypothesis is that by training for more epochs, we are able to obtain a better model.



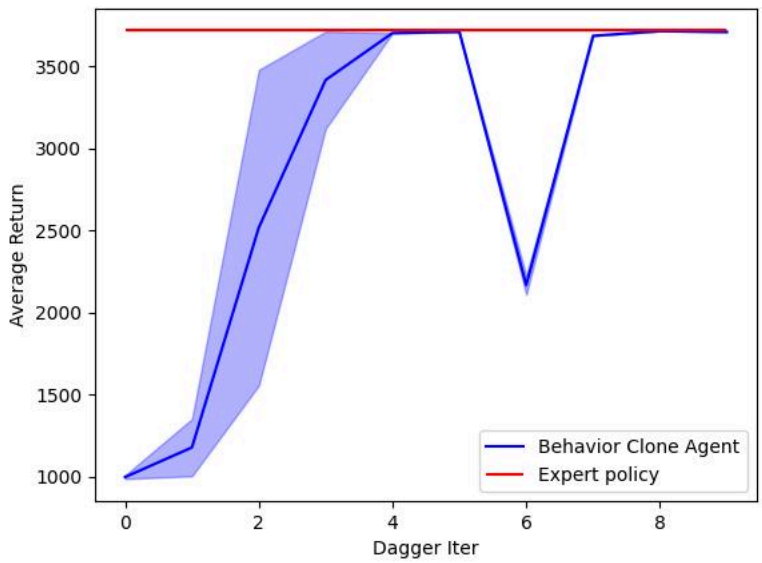

For Hopper, by training 25000 steps, we are able to achieve 2641 eval average return, which is 71% of the expert policy.

# 4 Dagger

Learning curve for HalfCheetah (default setting)




Learning curve for Hopper (default setting)


3500
3000
2500
2000
1500
1000
Average Return
0
2
4
6
8
Dagger Iter
Behavior Clone Agent
Expert policy